US012418138B2

(12) United States Patent
Li et al.

(10) Patent No.: US 12,418,138 B2
(45) Date of Patent: Sep. 16, 2025

(54) **ELECTRICAL CONNECTOR WITH WATERPROOF MEMBER THEREIN**

(71) Applicant: **Advanced Connectek Inc.**, New Taipei (TW)

(72) Inventors: **Yubin Li**, New Taipei (TW); **Yao Mei Wang**, New Taipei (TW); **Guan Liao**, New Taipei (TW); **Pin-Yuan Hou**, New Taipei (TW); **Shu-Fen Wang**, New Taipei (TW)

(73) Assignee: **Advanced Connectek Inc.**, New Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 450 days.

(21) Appl. No.: **17/993,876**

(22) Filed: **Nov. 24, 2022**

(65) **Prior Publication Data**

US 2023/0170646 A1 Jun. 1, 2023

(30) **Foreign Application Priority Data**

Nov. 26, 2021 (CN) 202111420123.7

(51) **Int. Cl.**

H01R 13/52	(2006.01)
H01R 4/02	(2006.01)
H01R 13/502	(2006.01)
H01R 24/60	(2011.01)

(52) **U.S. Cl.**

CPC ***H01R 13/5216*** (2013.01); ***H01R 4/023*** (2013.01); ***H01R 13/502*** (2013.01); ***H01R 24/60*** (2013.01)

(58) **Field of Classification Search**

CPC H01R 13/5216; H01R 13/523; H01R 13/502; H01R 24/60

USPC 439/519, 520, 521

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,761,988	B1 *	9/2017	Lin	H01R 13/5216
9,899,779	B1 *	2/2018	Lin	H01R 13/5202
12,088,032	B2 *	9/2024	Yang	H01R 13/405
2018/0097298	A1 *	4/2018	Ju	H01R 13/6591
2021/0111508	A1 *	4/2021	Gu	H01R 43/005

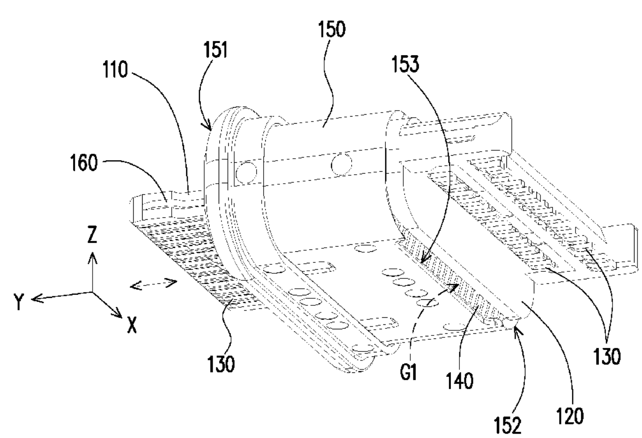
* cited by examiner

Primary Examiner — Abdullah A Riyami
Assistant Examiner — Amara Anderson
(74) *Attorney, Agent, or Firm* — JCIPRNET

(57) **ABSTRACT**

An electrical connector including a plurality of terminals, a first insulating body, a second insulating body, a shell, and a waterproof member is provided. The terminals are respectively retained in the first insulating body and the second insulating body. The shell is sheathed to the first insulating body and the second insulating body. The shell has a front edge and a rear edge opposite to each other along an insertion direction. The front edge is used for mating another electrical connector. At least one opening is formed by the rear edge and the second insulating body. The first insulating body and the second insulating body are separated from each other. A space is formed by the first insulating body, the second insulating body, and the shell. The space is communicated with an external environment via the at least one opening. The waterproof member is filled into the space.

17 Claims, 6 Drawing Sheets



100

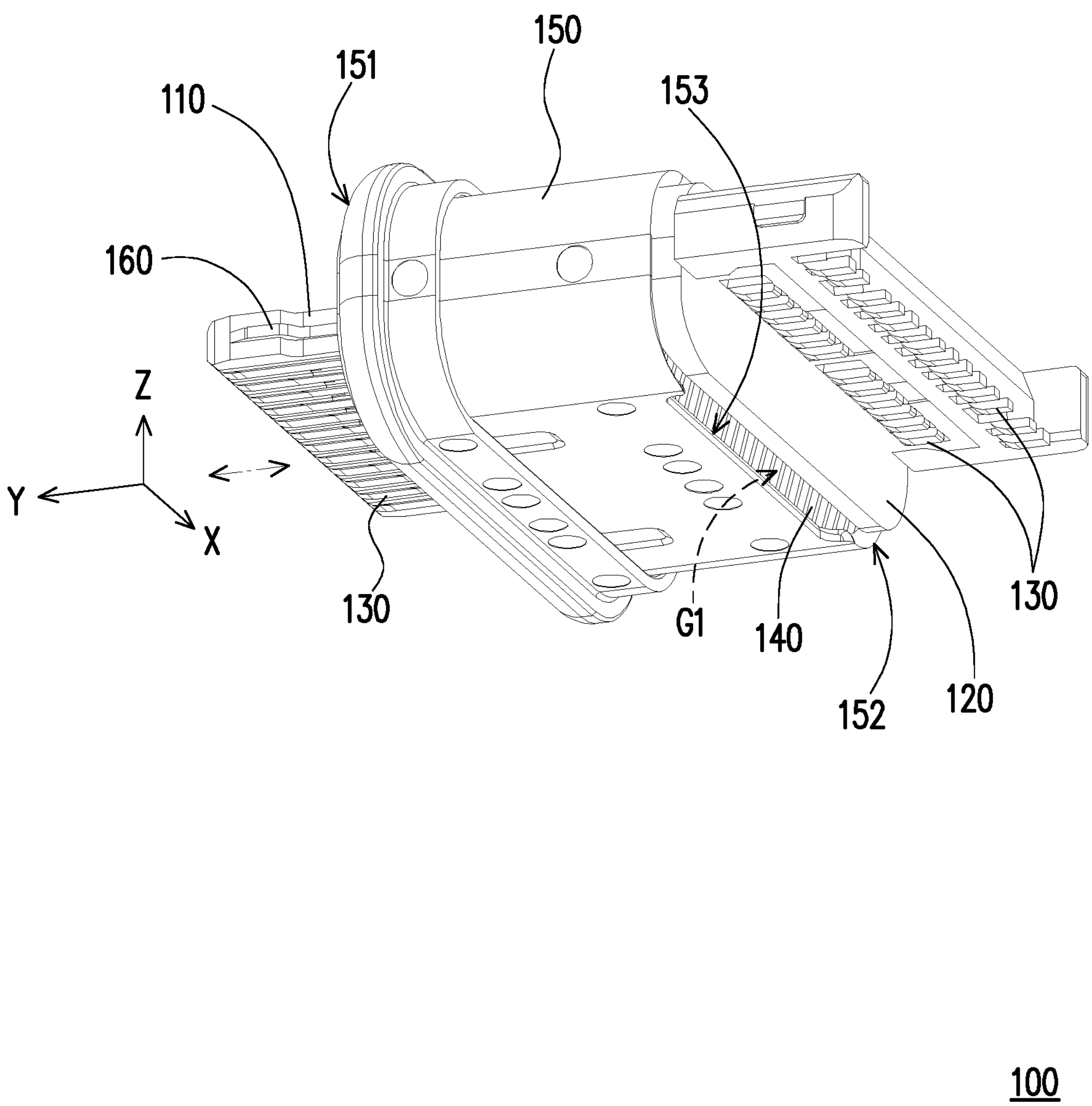


FIG. 1

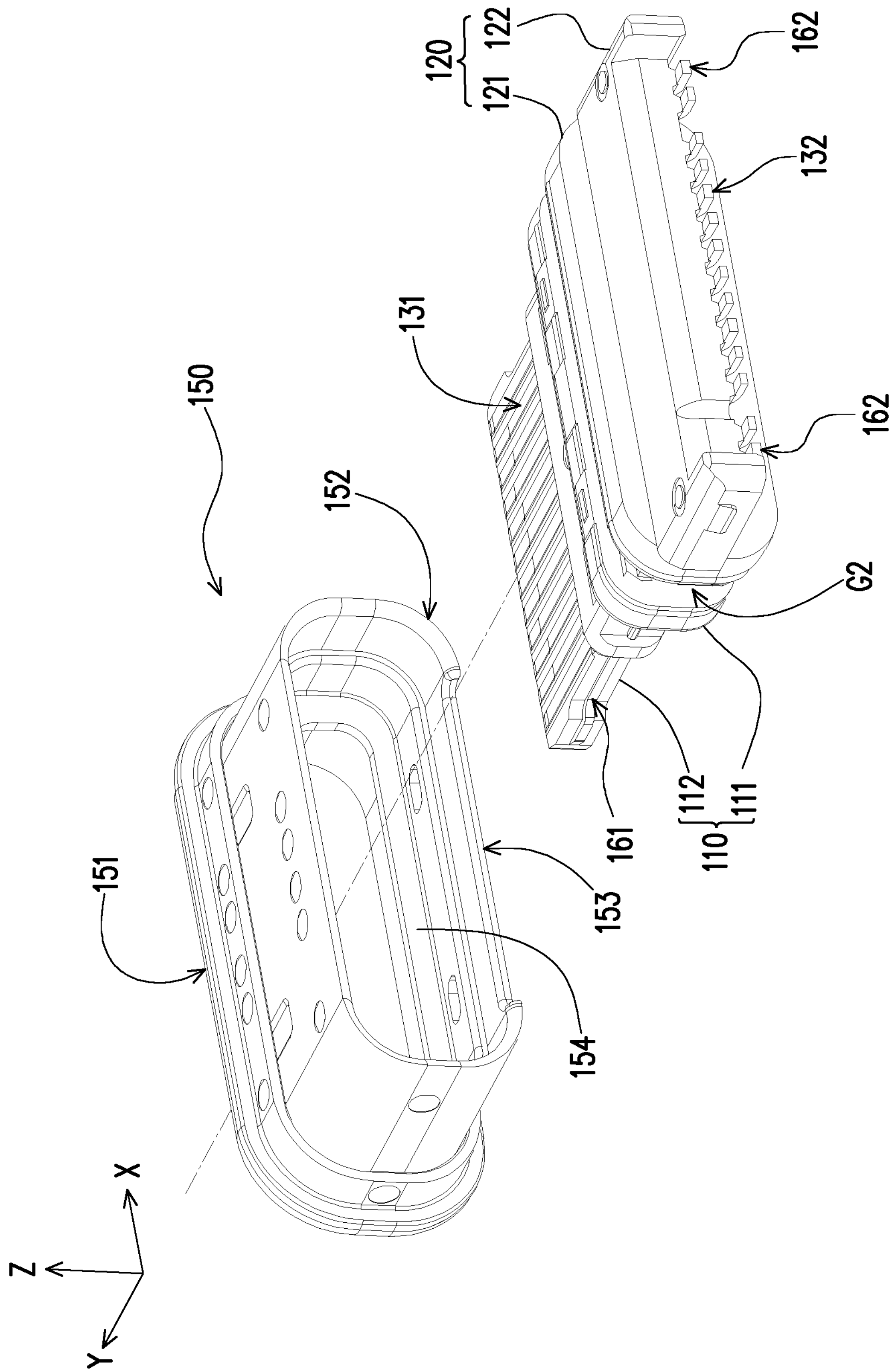

120
121
122
162
132
131
150
152
162
G2
161
112
111
110
151
153
154
X
Z
Y

FIG. 2

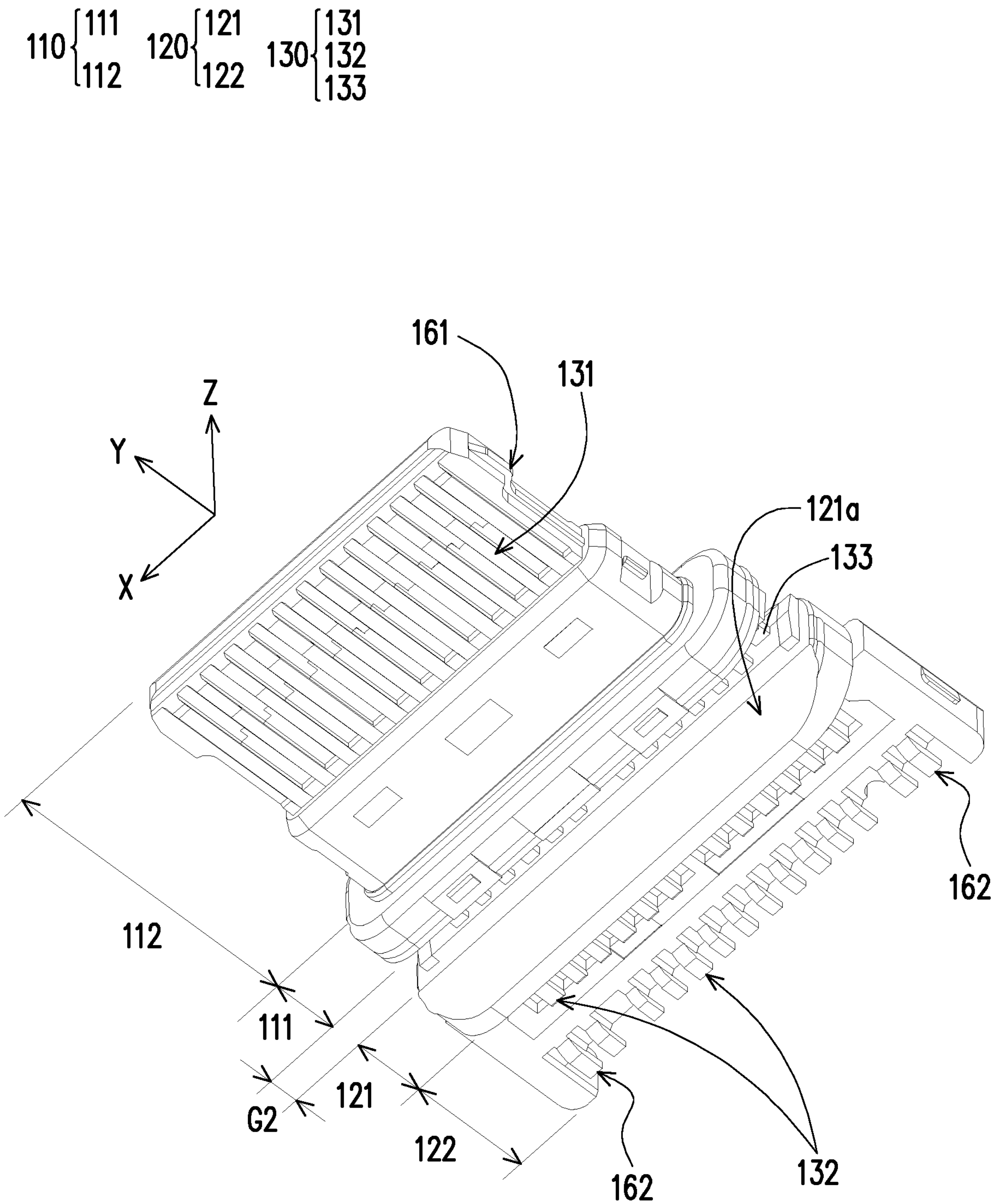

110 111 112
120 121 122
130 131 132 133
161
131
Z
Y
X
121a
133
162
112
111
121
G2
122
162
132

FIG. 3

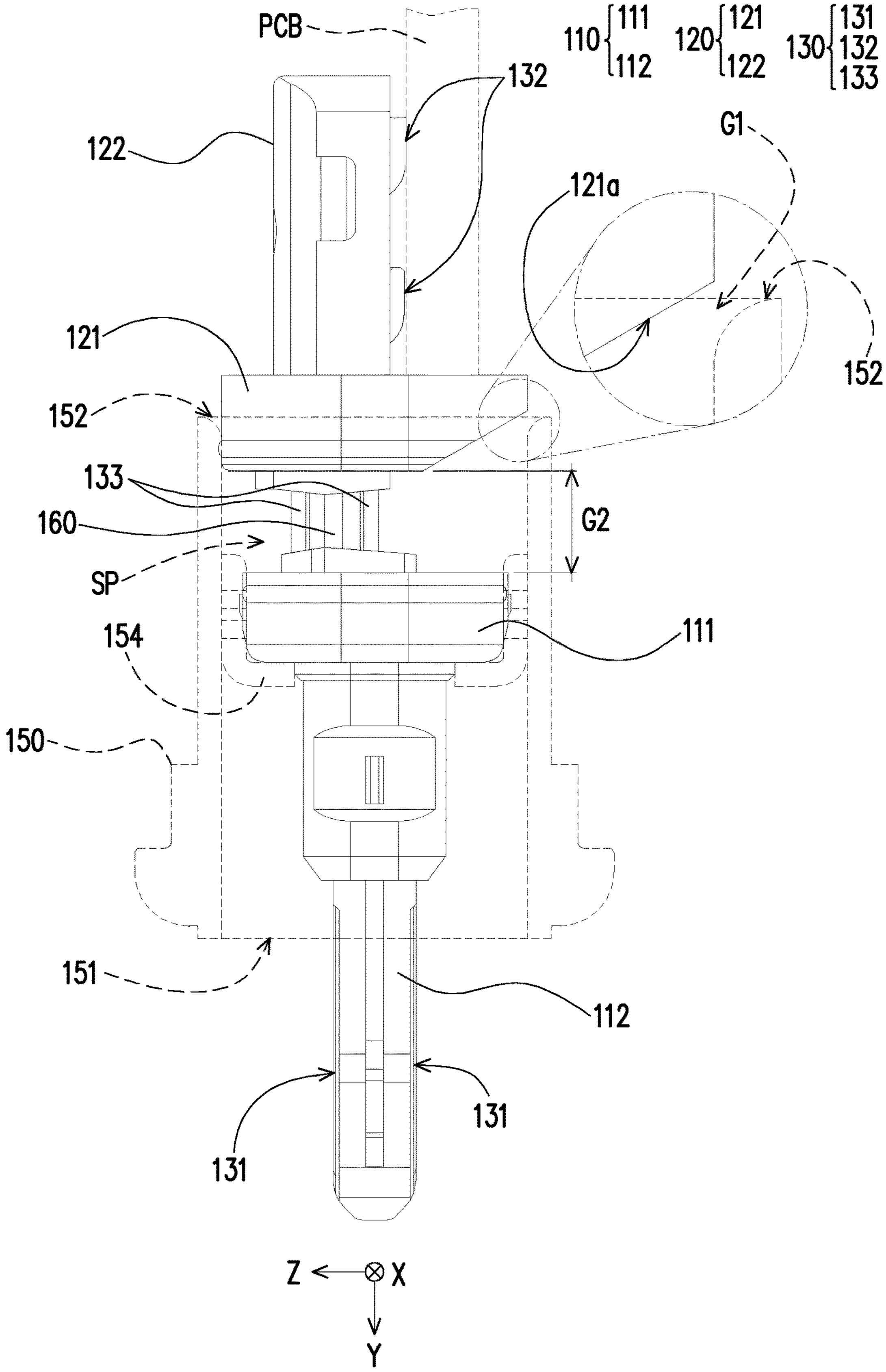

PCB
110
111
112
120
121
122
130
131
132
133
132
122
G1
121a
121
152
152
133
160
G2
SP
111
154
150
151
112
131
131
Z
X
Y

FIG. 4

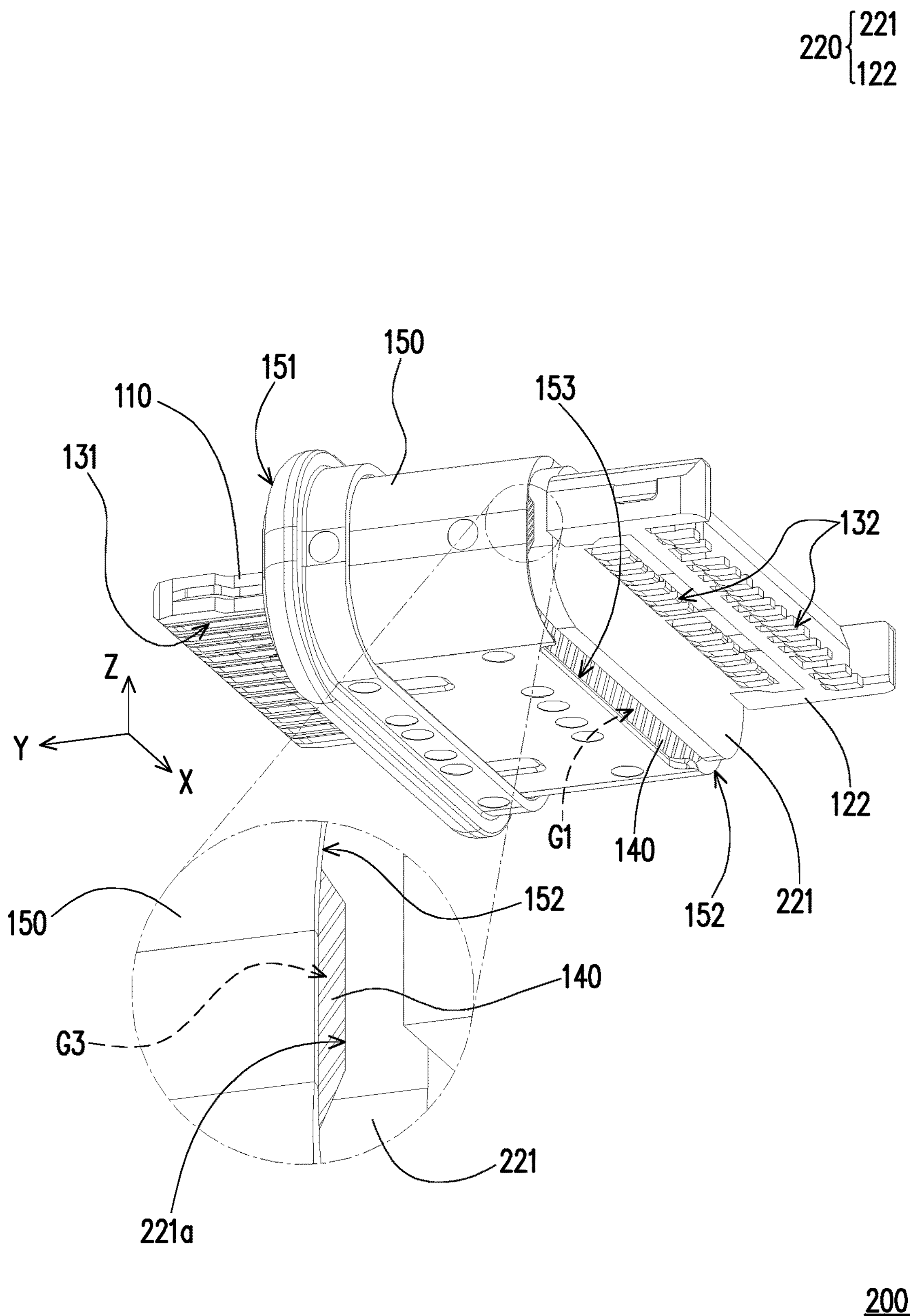

220
221
122
150
151
110
131
153
132
Z
Y
X
G1
140
122
152
221
150
152
140
G3
221
221a
200

FIG. 5

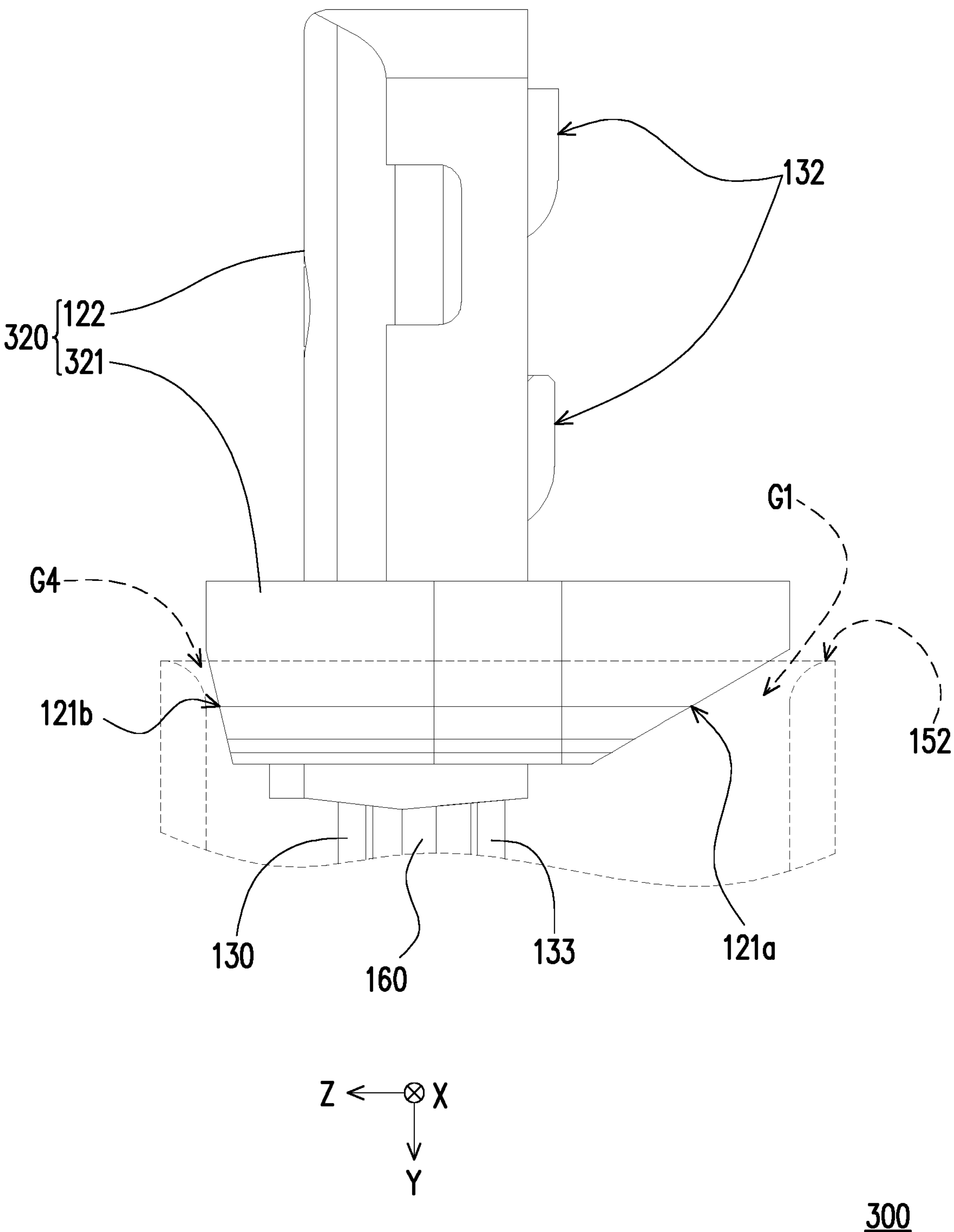


FIG. 6

ELECTRICAL CONNECTOR WITH WATERPROOF MEMBER THEREIN

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of China application serial no. 202111420123.7, filed on Nov. 26, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

The invention relates to an electrical connector.

Description of Related Art

The waterproof effect of the existing electrical connector products mostly adopts external sealing. That is, a waterproof rubber ring is disposed between the electrical connector and the body of the applied product in order to prevent water from entering the body from the external environment from the seam between the electrical connector and the body.

As mentioned above, the prior art does not provide any waterproof means for the internal structure of the electrical connector. For example, generally, when the electrical connector is combined with the internal circuit board of the body, it is often necessary to solder the two at high temperature. At this time, the insulating bodies and the terminals of the electrical connector are readily deformed in different manners due to the difference in specific heat, thereby generating a gap. As a result, if the waterproof mechanism of the internal structure of the electrical connector is neglected, water from the external environment readily enters the electrical connector, such that the waterproof effect is lost.

SUMMARY OF THE INVENTION

The invention provides an electrical connector that provides a corresponding waterproof mechanism from the internal structure of the electrical connector, and particularly provides a waterproof mechanism between insulating bodies and terminals.

An electrical connector of the invention includes a plurality of terminals, a first insulating body, a second insulating body, a shell, and a waterproof member. The terminals are respectively retained in the first insulating body and the second insulating body. The shell is sheathed to the first insulating body and the second insulating body. The shell has a front edge and a rear edge opposite to each other along an insertion direction. The front edge is used for mating another electrical connector. At least one opening is formed by the rear edge and the second insulating body. The first insulating body and the second insulating body are separated from each other. A space is formed by the first insulating body, the second insulating body, and the shell. The space is communicated with an external environment via the at least one opening. The waterproof member is filled into the space.

An electrical connector of the invention includes a plurality of terminals, a first insulating body, a second insulating body, a shell, and a waterproof member. The terminals are respectively retained in the first insulating body and the second insulating body along an insertion direction. A gap is between the first insulating body and the second insulating body, and a part of each terminals is exposed in the gap.

Preferably, the waterproof member is a waterproof adhesive injected through the at least one opening and filled in the space.

Preferably, the waterproof member is a waterproof adhesive injected via the at least one opening and fully filled in the space and the gap to enclose the exposed part of each terminal.

Preferably, at least one of the rear edge and the second insulating body has at least one notch at a portion where the rear edge and the second insulating body are adjacent to each other to form the at least one opening.

Preferably, the second insulating body has an inclined surface inclined from the at least one opening toward the space.

Preferably, the first insulating body is disposed in the shell and located between the front edge and the rear edge.

Preferably, the first insulating body has a first base portion and a tongue portion, the first base portion is abutted against a retaining wall in the shell, the tongue portion passes through and is protruded beyond the shell, and the first base portion is located between the tongue portion and the second insulating body.

Preferably, each of the plurality of terminals has a contact portion for mating with the other electrical connector. The contact portion exposes the first insulating body from the tongue portion.

Preferably, the second insulating body has a second base portion and a tail portion, the first base portion and the second base portion maintain a gap along the insertion direction, and the second base portion is located between the first base portion and the tail portion.

Preferably, each of the plurality of terminals has a soldering portion for soldering on a circuit board, and the soldering portion exposes the second insulating body from the tail portion.

Preferably, the second base portion and the rear edge of the shell are adjacent to each other and the at least one opening is therebetween.

Preferably, the second base portion is seated on a plane, and the insertion direction is a normal of the plane. An orthographic projection area of the waterproof member on the plane is less than that of the second base portion, so that the waterproof member is shielded by the second base portion in the insertion direction.

Preferably, the at least one opening includes a plurality of openings disposed around the insertion direction.

Based on the above, the electrical connector is respectively combined to the terminals via the first insulating body and the second insulating body separated from each other, so as to allow the shell, the first insulating body, and the second insulating body to form the space and allow the second insulating body and the rear edge of the shell to form the opening and further allow the space to be communicated with the external environment via the opening when the shell is sheathed on the first insulating body and the second insulating body. In this way, when the waterproof member is filled in the space, it is equivalent to having the waterproof member form a waterproof structure between the shell, the first insulating body, and the second insulating body separated from each other. In addition to the waterproof mechanism between the shell and the first insulating body and the second insulating body, even if there is a gap between the terminals and the insulating bodies for the above reasons, the gap may still be effectively blocked. That is, a perfect waterproof mechanism is provided from the internal structure of the electrical connector.

To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

FIG. **1** is a schematic diagram of an electrical connector according to an embodiment of the invention.

FIG. **2** is an exploded view of the electrical connector of FIG. **1**.

FIG. **3** shows some members of FIG. **2** from another perspective.

FIG. **4** is a side view of the electrical connector of FIG. **1**.

FIG. **5** is a schematic diagram of an electrical connector of another embodiment of the invention.

FIG. **6** is a side view of an electrical connector of another embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

FIG. **1** is a schematic diagram of an electrical connector according to an embodiment of the invention. FIG. **2** is an exploded view of the electrical connector of FIG. **1**. FIG. **3** shows some members of FIG. **2** from another perspective. Cartesian coordinates X-Y-Z are also provided here to facilitate member description. Referring to all of FIG. **1** to FIG. **3**, in the present embodiment, an electrical connector **100** is used for mating with another electrical connector (not shown) along an insertion direction (double arrows shown in FIG. **1**). As shown in FIG. **1**, and the electrical connector **100** of the present embodiment is mated to the other electrical connector along the Y-axis, and the Y-axis is the insertion direction of the electrical connector **100**.

The electrical connector **100** includes a plurality of terminals **130**, a first insulating body **110**, a second insulating body **120**, a shell **150**, and a waterproof member **140**. The first insulating body **110** and the second insulating body **120** are respectively combined to the terminals **130**. In other words, there exists a gap or a spacing G2 between the first insulating body **110** and the second insulating body **120**. The shell **150** is sheathed on the first insulating body **110** and the second insulating body **120**. The shell **150** has a front edge **151** and a rear edge **152** opposite to each other along the insertion direction. The front edge **151** is used for mating with another electrical connector, and the rear edge **152** and the second insulating body **120** form at least one opening G1. The first insulating body **110** and the second insulating body **120** are separated from each other.

FIG. **4** is a side view of the electrical connector of FIG. **1**, and the waterproof member **140** shown in FIG. **1** is omitted here to facilitate the identification of internal structural features formed by the shell **150**, the first insulating body **110**, and the second insulating body **120**. Please refer to FIG. **2** to FIG. **4** at the same time, in the present embodiment, after the shell **150** is sheathed on the first insulating body **110** and the second insulating body **120**, the first insulating body **110**, the second insulating body **120**, and the shell **150** form a space SP, and the space SP is communicated with the external environment via the opening G1. The waterproof member **140** shown in FIG. **1** is filled in the space SP.

Further, the first insulating body **110** is disposed in the shell **150** and located between the front edge **151** and the rear edge **152**. At least one of the rear edge **152** and the second insulating body **120** has at least one notch **153** at a portion where the rear edge **152** and the second insulating body **120** are adjacent to each other to form the at least one opening G1. Here, the notch **153** formed by the rear edge **152** of the shell **150** is used as an example. Moreover, the first insulating body **110** has a first base portion **111** and a tongue portion **112**, the tongue portion **112** is extended beyond the first base portion **111**, and each of the terminals **130** has a contact portion **131**, a connection portion **133**, and a soldering portion **132**. The contact portion **131** and the soldering portion **132** are opposite to each other, wherein the contact portion **131** exposes the first insulating body **110** from the upper and lower surfaces of the tongue portion **112** and is arranged in double rows along the Z-axis in order to facilitate mating with the other electrical connector. The connection portion **133** is connected between the contact portion **131** and the soldering portion **132**, wherein a part of connection portion **133** is exposed in the gap G2.

In addition, the first base portion **111** is abutted against the retaining wall **154** in the shell **150**, the tongue portion **112** passes through and is protruded beyond the shell **150**, and the first base portion **111** is located between the tongue portion **112** and the second insulating body **120**.

In contrast, the second insulating body **120** has a second base portion **121** and a tail portion **122**, the first base portion **111** and the second base portion **121** maintain the gap or the spacing G2 along the insertion direction, and the second base portion **121** is located between the first base portion **111** and the tail portion **122**. The second base portion **121** and the rear edge **152** of the shell **150** are adjacent to each other and the at least one opening G1 is therebetween. As shown in FIG. **2** and FIG. **3**, the soldering portion **132** of each of the terminals **130** exposes the second insulating body **120** from the tail portion **122**, and is arranged in double rows along the Y-axis. The soldering portion **132** is substantially coplanar (equivalent to being located on the X-Y plane) to facilitate soldering to pads (not shown) on a circuit board PCB.

Moreover, the electrical connector **100** further includes a shielding sheet **160** combined with the first insulating body **110** and the second insulating body **120** together with the terminal **130** via an insert molding technique, wherein the shielding sheet **160** is located between the terminal **130** arranged in double rows, so as to prevent the high-speed signals of the terminal **130** from interfering with each other. Moreover, the shielding sheet **160** has a latching terminal **161** and a grounding terminal **162** opposite to each other, wherein the latching terminal **161** exposes the first insulating body **110** from the tongue portion **112**, and the grounding terminal **162** and one row of the soldering portions **132** are in the same row and coplanar, so as to facilitate soldering to the circuit board PCB and provide a grounding effect.

It should be mentioned that, the shielding sheet **160** and a portion of the terminal **130** are exposed between the first insulating body **110** and the second insulating body **120** and located in the space SP. However, since the waterproof member **140** may be fully filled in the space SP and the gap or the spacing G2 so as to enclose the exposed part of each terminal **130**, and the waterproof effect between the members may be effectively produced.

It should be mentioned that, the arrangement of the terminal **130** and the shielding sheet **160** is the same as that disclosed in the prior art electrical connector. Therefore, the above-mentioned and unmentioned parts of the present embodiment may be known from the prior art and are not repeated herein.

As shown in FIG. **4**, the assembled shell **150**, first insulating body **110**, and second insulating body **120** form the space SP, and the space SP of the present embodiment is communicated with the external environment via the opening G**1**, and after the above assembly, the operator may further inject a waterproof adhesive into the space SP via the opening G**1**. After the waterproof adhesive is cured, the waterproof member **140** is formed. As shown in FIG. **1**, since the opening G**1** is in the shape of a long strip, sufficient space and position are provided for the operator to inject the adhesive, and also the discharge of the gas in the space SP is facilitated. Furthermore, as shown in FIG. **4**, the second base portion **121** of the second insulating body **120** of the present embodiment also has an inclined surface **121***a* inclined from the opening G**1** toward the space SP, thus facilitating to provide a guiding effect for the injected waterproof adhesive.

As shown in FIG. **4**, the waterproof adhesive filled in the space SP (which becomes the waterproof member **140** after curing) is matched with the first insulating body **110** and the second insulating body **120** separated from each other along the insertion direction. Therefore, a waterproof mechanism may be effectively provided between the shell **150** and the first insulating body **110** and the second insulating body **120**, and between the terminal **130** and the first insulating body **110** and the second insulating body **120**, in order to achieve the desired waterproof effect from the internal structure of the electrical connector **100**.

Moreover, the second base portion **121** is substantially located on the X-Z plane, and the insertion direction (the Y-axis) is the normal of the X-Z plane. The orthographic projection area of the waterproof member **140** on the X-Z plane is less than that of the second base portion **121**. Therefore, the waterproof member **140** is in a state of being shielded by the second base portion **121** in the insertion direction. As shown in FIG. **4**, when the user looks at the electrical connector **100** along the Y-axis from the circuit board PCB, only the second base portion **121** is seen and the waterproof member **140** is not seen as exposed.

FIG. **5** is a schematic diagram of an electrical connector of another embodiment of the invention. Please refer to FIG. **5** and compare with FIG. **1**. Different from the previous embodiment, in an electrical connector **200** of the present embodiment, in addition to the same opening G1 as the previous embodiments, the second base portion **221** of the second insulating body **220** of the present embodiment also has another notch **221***a* to form another opening G3 with the rear edge **152** of the shell **150**. Accordingly, when the operator performs the adhesive filling operation as described above, the opening G1 may be used for adhesive filling, and the opening G**3** may be used for exhausting. In FIG. **5**, the same reference numerals as the previous embodiment or the same member features as the previous embodiment are described in the previous embodiment, and are not repeated herein.

It may be known from the embodiment shown in FIG. **5** and the embodiment shown in FIG. **1** to FIG. **4** that the quantity and position of the openings are not limited in the present application. That is, one or a plurality of openings G1 (or G3) may be disposed at a portion where the rear edge **152** of the shell **150** and the second base portion **121** (or **221**) of the second insulating body **120** (or **220**) are adjacent to each other and at a position where the one or a plurality of openings G1 (or G3) are disposed around the insertion direction.

FIG. **6** is a side view of an electrical connector of another embodiment of the invention. Please refer to FIG. **6** and compare with FIG. **4**. As mentioned above, the quantity and position of the openings are not limited in the present application. Therefore, in an electrical connector **300** of the present embodiment, in addition to the same opening G1 as in the previous embodiments, the second base portion **321** of the second insulating body **320** of the present embodiment also has another inclined surface **121***b* inclined from another opening G**4** toward the space SP, and the inclined surface **121***b* and the opening G**4** and the inclined surface **121***a* and the opening G**1** are opposite to each other along the Z-axis. Relevant structural features produce the same effects as those of the above embodiments and are not repeated herein.

Based on the above, in the embodiments of the invention, the electrical connector is respectively combined to the terminals via the first insulating body and the second insulating body separated from each other, so as to allow the shell, the first insulating body, and the second insulating body to form the space and allow the second insulating body and the rear edge of the shell to form the opening and further allow the opening to be used to communicate with the space and the external environment when the shell is sheathed on the first insulating body and the second insulating body.

In particular, in different embodiments, desired openings may be formed at the second insulating body and the rear edge of the shell according to the adhesive pouring operation and the exhaust requirements.

In this way, when the waterproof member is filled in the space, it is equivalent to having the waterproof member form a waterproof structure between the shell, the first insulating body, and the second insulating body separated from each other. In addition to the waterproof mechanism between the shell and the first insulating body and the second insulating body, even if there is a gap between the terminals and the insulating bodies for the above reasons, the gap may still be effectively blocked. That is, a perfect waterproof mechanism is provided from the internal structure of the electrical connector.

It will be apparent to those skilled in the art that various modifications and variations may be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

What is claimed is:

1. An electrical connector, comprising:

a plurality of terminals;

a first insulating body and a second insulating body, wherein the plurality of terminals are respectively retained in the first insulating body and the second insulating body;

a shell sheathed to the first insulating body and the second insulating body, wherein the shell has a front edge and a rear edge opposite to each other along an insertion direction, the front edge is used for mating another electrical connector, at least one opening is formed by the rear edge and the second insulating body, the first insulating body and the second insulating body are separated from each other, a space is formed by the first insulating body, the second insulating body, and the shell, and the space is communicated with an external environment via the at least one opening; and a waterproof member filled into the space,

wherein the waterproof member is a waterproof adhesive injected via the at least one opening and filled in the space,

wherein the second insulating body has an inclined surface inclined from the at least one opening toward the space to guide the waterproof adhesive injected from the at least one opening to flow toward the space.

2. The electrical connector of claim **1**, wherein at least one of the rear edge and the second insulating body has at least one notch at a portion where the rear edge and the second insulating body are adjacent to each other to form the at least one opening.

3. The electrical connector of claim **1**, wherein the first insulating body is disposed in the shell and located between the front edge and the rear edge.

4. The electrical connector of claim **1**, wherein the first insulating body has a first base portion and a tongue portion, the first base portion is abutted against a retaining wall in the shell, the tongue portion passes through and is protruded beyond the shell, and the first base portion is located between the tongue portion and the second insulating body.

5. The electrical connector of claim **4**, wherein the plurality of terminals each have a contact portion for mating with the other electrical connector, and the contact portion respectively exposes on an upper surface and a lower surface of the tongue portion of the first insulating body.

6. The electrical connector of claim **4**, wherein the second insulating body has a second base portion and a tail portion, the first base portion and the second base portion maintain a spacing along the insertion direction, and the second base portion is located between the first base portion and the tail portion.

7. The electrical connector of claim **6**, wherein each of the plurality of terminals has a soldering portion for soldering on a circuit board, and the soldering portion exposes the second insulating body from the tail portion.

8. The electrical connector of claim **6**, wherein the second base portion and the rear edge of the shell are adjacent to each other and the at least one opening is therebetween.

9. The electrical connector of claim **6**, wherein the second base portion is seated on a plane, the insertion direction is a normal of the plane, and an orthographic projection area of the waterproof member on the plane is less than that of the second base portion, so that the waterproof member is shielded by the second base portion in the insertion direction.

10. An electrical connector, comprising:

a plurality of terminals;

a first insulating body and a second insulating body, wherein the plurality of terminals are respectively retained in the first insulating body and the second insulating body along an insertion direction, a gap is between the first insulating body and the second insulating body, and a part of each terminal is exposed in the gap;

a shell sheathed to the first insulating body and the second insulating body, wherein the shell has a front edge and a rear edge opposite to each other along the insertion direction, the front edge is used for mating another electrical connector, at least one opening is formed by the rear edge and the second insulating body, the first insulating body and the second insulating body are separated from each other by the gap, a space is formed by the first insulating body, the second insulating body, and the shell, and the space is communicated with an external environment via the at least one opening; and

a waterproof member filled into the space and the gap,

wherein the waterproof member is a waterproof adhesive injected via the at least one opening and fully filled in the space and the gap to enclose the exposed part of each terminal,

wherein the second insulating body has an inclined surface inclined from the at least one opening toward the space to guide the waterproof adhesive injected from the at least one opening to flow toward the space.

11. The electrical connector of claim **10**, wherein at least one of the rear edge and the second insulating body has at least one notch at a portion where the rear edge and the second insulating body are adjacent to each other to form the at least one opening.

12. The electrical connector of claim **10**, wherein the first insulating body is disposed in the shell and located between the front edge and the rear edge.

13. The electrical connector of claim **10**, wherein the first insulating body has a first base portion and a tongue portion, the first base portion is abutted against a retaining wall in the shell, the tongue portion passes through and is protruded beyond the shell, and the first base portion is located between the tongue portion and the second insulating body.

14. The electrical connector of claim **13**, wherein the plurality of terminals each have a contact portion for mating with the other electrical connector, and the contact portion respectively exposes on an upper surface and a lower surface of the tongue portion of the first insulating body.

15. The electrical connector of claim **13**, wherein the second insulating body has a second base portion and a tail portion, the first base portion and the second base portion maintain the gap along the insertion direction, and the second base portion is located between the first base portion and the tail portion.

16. The electrical connector of claim **15**, wherein the second base portion is seated on a plane, the insertion direction is a normal of the plane, and an orthographic projection area of the waterproof member on the plane is less than that of the second base portion, so that the waterproof member is shielded by the second base portion in the insertion direction.

17. An electrical connector, comprising:

a plurality of terminals;

a first insulating body and a second insulating body, wherein the plurality of terminals are respectively retained in the first insulating body and the second insulating body;

a shell sheathed to the first insulating body and the second insulating body, wherein the shell has a front edge and a rear edge opposite to each other along an insertion direction, the front edge is used for mating another electrical connector, at least one opening is formed by the rear edge and the second insulating body, the first insulating body and the second insulating body are separated from each other, a space is formed by the first insulating body, the second insulating body, and the shell, and the space is communicated with an external environment via the at least one opening; and

a waterproof member filled into the space,

wherein the first insulating body has a first base portion and a tongue portion, the first base portion is abutted against a retaining wall in the shell, the tongue portion passes through and is protruded beyond the shell, and the first base portion is located between the tongue portion and the second insulating body,

wherein the second insulating body has a second base portion and a tail portion, the first base portion and the second base portion maintain a spacing along the insertion direction, and the second base portion is located between the first base portion and the tail portion,

wherein the second base portion is seated on a plane, the insertion direction is a normal of the plane, and an orthographic projection area of the waterproof member on the plane is less than that of the second base portion, so that the waterproof member is shielded by the second base portion in the insertion direction.

* * * * *